Benchmark-Estimand Alignment

Benchmark Family (Representative Examples)	Chain Observability θ_{obs} ($C^* \subseteq L_N$)	Conditional Utility (Selection) θ_{util} ($C^* \subseteq E_K$)	Exposure (Ordering) θ_{exp} ($\text{rank}_\pi(C^*) \leq r$)	Fusion Reliability θ_{fus} (Compose C^*)	Causal Faithfulness θ_{faith} ($C^* \rightarrow \hat{a}$)	Primary Purpose
Supporting-fact QA *MuSiQue, 2WikiMultihopQA, HotpotQA*	●	◐	○	●	○	Test whether all supporting facts can be retrieved and combined correctly.
Shortcut-resistant QA *StrategyQA, Bamboogle, NQ-Swapped*	●	◐	●	◐	●	Discourage spurious heuristics and stress genuine multi-hop reasoning.
Path / Process Benchmarks *ProofWriter, GraphQ, LogicNLG*	◐	●	●	●	◐	Evaluate correctness of intermediate reasoning steps and evidence chaining.
Heterogeneous Evidence QA *HybridQA, xGQA, 2WikiMQA*	●	◐	◐	●	◐	Require reasoning over multiple evidence types such as text, tables, and KGs.
Long-context / Needle-in-a-Haystack *LongBench, NovelHopQA, InfiniteBench*	●	◐	●	◐	○	Stress long-context recall and early exposure of key evidence.
Fact Verification / NLI *FEVER, HoVer, EX-FEVER*	◐	◐	◐	◐	●	Measure whether claims are truly supported by retrieved evidence.
RAG-specific Diagnostics *RAGChecker, RGB, ARES*	●	●	◐	◐	●	Diagnose retriever-reader interaction and faithfulness issues in RAG.
Counterfactual / Intervention *CofCA, MEDR, CF-RAG*	◐	◐	◐	◐	●	Validate causal dependence by counterfactual perturbation or intervention.
Agentic / Tool-use Benchmarks *WebWalker, GAIA, BrowseComp*	●	●	●	◐	◐	Evaluate multi-step search, tool use, and adaptive retrieval policies.

Legend ● Primary alignment ◐ Secondary alignment ○ Weak alignment

Interpretation

Read each row as which latent-chain failures a benchmark most exposes. No single benchmark covers all five estimands; comprehensive evaluation requires a diverse benchmark suite.

Fig. 9. Alignment of benchmark families with the five latent-chain estimands. Different benchmarks stress different failure modes. Grouping benchmarks by the estimands they probe is more informative than grouping only by domain.